

Verification of whether the final matrix is in rref

1. Row 1, 2 and 3 are all non-zero rows. $\therefore$ The condition that all non-zero rows must be above the zero rows is vacuously True.
 2. Row 1 Leading entry = 1 in Column 1
Row 2 Leading entry = 1 in Column 2
Row 3 Leading entry = 1 in Column 3
 $\therefore$ The condition that for each row the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.
 3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all positions below the leading entry position must contain 0 is True.
 4. The condition that each leading entry must be 1 is True.
 5. Row 1 Column 3 = Row 2 Column 3 = 0
Row 1 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all positions above the leading entry position must contain 0 is True.
- $\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ is in rref. Apart from the last column, all columns contain pivots. By defn, there are no free variables.
- $\therefore$ By Theorem 2, the matrix equation $Ax=0$ has a unique solution.
- $\therefore$ The columns of matrix A are linearly independent $\square$

Sets of One or Two Vectors

A set containing only one vector — say, v — is linearly independent if and only if v is not the zero vector. This is because the vector equation $\gamma v = 0$ has only the trivial solution when $v \neq 0$. The zero vector is linearly dependent because $\gamma 0 = 0$ has many nontrivial solutions.